Resistivity of Nichrome Wire

Experiment 7 • PHYS 210 X01A

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Objective

1. To verify the relationship between the DC electrical resistance R (Ω) and the geometric dimensions of metal wire of cylindrical cross section: the length L and the diameter d.

2. To experimentally measure the resistivity ρ (Ω m) of Nichrome wire and compare with the accepted value.

Theory

A longer wire has more resistance. A thicker wire has less resistance because it has a larger cross-sectional area. For a uniform wire at a constant temperature:

$$R = \frac{\rho L}{A} \qquad A = \frac{\pi d^2}{4}$$

R is resistance (Ω), ρ is resistivity (Ω m), L is wire length (m), A is cross-sectional area (m^2), and d is diameter (m).

In Part A, we used the same wire diameter and changed the length. The graph of resistance against length should be a straight line, with slope $m_A = \rho/A$. In Part B, the length stayed the same and the wire diameter changed. The graph of resistance against inverse area should also be a straight line, with slope $m_B = \rho L$. Both lines should pass through zero, but we let Excel fit the measured points without forcing this.

$$\rho_A = m_A A \qquad \rho_B = \frac{m_B}{L}$$

Δ means uncertainty. When multiplying or dividing, we added the fractional uncertainties shown below. The slope uncertainty was half the difference between the maximum and minimum slopes.

$$\Delta m = \frac{m_{max} - m_{min}}{2} \qquad \frac{\Delta A}{A} = \frac{2\Delta d}{d}$$

$$\frac{\Delta \rho_A}{\rho_A} = \frac{\Delta m_A}{m_A} + \frac{2\Delta d}{d} \qquad \frac{\Delta \rho_B}{\rho_B} = \frac{\Delta m_B}{m_B} + \frac{\Delta L}{L}$$

Apparatus and procedure

We used Nichrome wire, a ruler, a Fluke multimeter with leads, and a Cenco resistance apparatus (catalogue 83406). We measured the resistance of #28 wire from 0.100 to 1.900 m, increasing the length by 0.100 m each time. We then measured six wire gauges, each with a length of 10.0 m.

We used the manual's AWG diameters to calculate the wire areas. We plotted the data with uncertainties in Excel and used the slopes to calculate resistivity.

Part A data and sample calculations

The #28 wire diameter was (0.3211 ± 0.0001) mm. The uncertainty in each length was ±0.001 m. We followed the lab manual's Fluke instructions to calculate resistance uncertainty: 0.9% of the reading plus 0.2 Ω.

L (m)	ΔL (m)	R (Ω)	ΔR (Ω)
0.100	0.001	1.9	0.2
0.200	0.001	3.3	0.2
0.300	0.001	5.2	0.2
0.400	0.001	6.2	0.3
0.500	0.001	7.6	0.3
0.600	0.001	9.1	0.3
0.700	0.001	10.4	0.3
0.800	0.001	11.9	0.3
0.900	0.001	13.4	0.3
1.000	0.001	14.9	0.3
1.100	0.001	16.3	0.3
1.200	0.001	17.9	0.4
1.300	0.001	19.2	0.4
1.400	0.001	20.7	0.4
1.500	0.001	22.4	0.4
1.600	0.001	23.8	0.4
1.700	0.001	25.1	0.4
1.800	0.001	26.5	0.4
1.900	0.001	27.9	0.5

Table 1. Length and resistance of #28 Nichrome wire.

$$\Delta R = 0.009(1.9\ \Omega) + 0.2\ \Omega = 0.2171\ \Omega \approx 0.2\ \Omega$$

$$d = 0.3211\ mm \times \frac{10^{-3}\ m}{1\ mm} = 3.211 \times 10^{-4}\ m$$

$$A = \frac{\pi(3.211 \times 10^{-4}\ m)^2}{4} = 8.097864 \times 10^{-8}\ m^2$$

$$\Delta A = A\frac{2\Delta d}{d} = (8.097864 \times 10^{-8}\ m^2)\frac{2 \times 0.0001\ mm}{0.3211\ mm}$$

$$\Delta A = 5.043827 \times 10^{-11}\ m^2 \approx 5 \times 10^{-11}\ m^2$$

Resistance increased from 1.9 Ω to 27.9 Ω as the wire got longer. The reading at 0.300 m was slightly above the line.

Part B data and sample calculations

Each wire was (10.0 ± 0.1) m long. We used the diameters in the manual's AWG table and assumed an uncertainty of ±0.0001 mm for each, based on the #28 example from class.

AWG	d (mm)	R (Ω)	ΔR (Ω)
20	0.8118	22.4	0.4
22	0.6438	36.2	0.5
24	0.5106	56.8	0.7
26	0.4049	98.5	1
28	0.3211	146.3	2
30*	0.2546	513.2	5

Table 2. Recorded diameters and resistances. *#30 was excluded from the fit.

AWG	A (m^2)	ΔA (m^2)	1/A (m^{-2})	Δ(1/A) (m^{-2})
20	5.175925e-07	1.275e-10	1.932022e+06	4.760e+02
22	3.255306e-07	1.011e-10	3.071908e+06	9.543e+02
24	2.047630e-07	8.020e-11	4.883695e+06	1.913e+03
26	1.287613e-07	6.360e-11	7.766307e+06	3.836e+03
28	8.097864e-08	5.044e-11	1.234894e+07	7.692e+03
30	5.091042e-08	3.999e-11	1.964234e+07	1.543e+04

Table 3. Calculated areas and inverse areas. Extra digits were kept during calculations.

For #20 wire:

$$A = \frac{\pi(0.8118 \times 10^{-3}\ m)^2}{4} = 5.175925 \times 10^{-7}\ m^2$$

$$\Delta A = (5.175925 \times 10^{-7}\ m^2)\frac{2 \times 0.0001\ mm}{0.8118\ mm} = 1.275172 \times 10^{-10}\ m^2$$

$$\frac{1}{A} = \frac{1}{5.175925 \times 10^{-7}\ m^2} = 1.932022 \times 10^{6}\ m^{-2}$$

$$\Delta\left(\frac{1}{A}\right) = \frac{1}{A}\frac{\Delta A}{A} = 475.985\ m^{-2}$$

$$\Delta R = 0.009(22.4\ \Omega) + 0.2\ \Omega = 0.4016\ \Omega \approx 0.4\ \Omega$$

We excluded the #30 reading of 513.2 Ω from the Part B graph and resistivity calculation because it was far above the trend of the other five readings. It is kept in the tables as recorded. The cause of this unusual reading was not confirmed.

Resistivity calculations

Part A

The slope in Figure 1 was 14.49 Ω/m, which matched the expected slope for #28 wire.

Fit quantity	Value
Best-fit slope	14.494737 Ω/m
Maximum slope	15.170912 Ω/m
Minimum slope	13.818561 Ω/m
Best-fit intercept	0.436842 Ω

$$\Delta m_A = \frac{(15.170912 - 13.818561)\ \Omega/m}{2} = 0.676176\ \Omega/m$$

$$m_A = (14.494737 \pm 0.676176)\ \Omega/m \approx (14.5 \pm 0.7)\ \Omega/m$$

$$\rho_A = (14.494737\ \Omega/m)(8.097864 \times 10^{-8}\ m^2) = 1.173764 \times 10^{-6}\ \Omega\ m$$

$$\Delta\rho_A = (1.173764 \times 10^{-6}\ \Omega\ m)\left[\frac{0.676176\ \Omega/m}{14.494737\ \Omega/m} + \frac{2(0.0001\ mm)}{0.3211\ mm}\right]$$

$$\Delta\rho_A = 0.055487 \times 10^{-6}\ \Omega\ m \qquad \rho_A = (1.17 \pm 0.06) \times 10^{-6}\ \Omega\ m$$

Part B

The fit in Figure 2 used five wire gauges (#20–#28), with #30 excluded.

Fit quantity	Value
Best-fit slope	1.206494×10^{-5} Ω m²
Maximum slope	1.304636×10^{-5} Ω m²
Minimum slope	1.108352×10^{-5} Ω m²
Best-fit intercept	−0.356561 Ω

$$\Delta m_B = \frac{(1.304636 - 1.108352) \times 10^{-5}\ \Omega\ m^2}{2} = 0.098142 \times 10^{-5}\ \Omega\ m^2$$

$$m_B = (1.206494 \pm 0.098142) \times 10^{-5}\ \Omega\ m^2 \approx (1.2 \pm 0.1) \times 10^{-5}\ \Omega\ m^2$$

$$\rho_B = \frac{1.206494 \times 10^{-5}\ \Omega\ m^2}{10.0\ m} = 1.206494 \times 10^{-6}\ \Omega\ m$$

$$\Delta\rho_B = (1.206494 \times 10^{-6}\ \Omega\ m)\left[\frac{0.098142 \times 10^{-5}\ \Omega\ m^2}{1.206494 \times 10^{-5}\ \Omega\ m^2} + \frac{0.1\ m}{10.0\ m}\right]$$

$$\Delta\rho_B = 0.110207 \times 10^{-6}\ \Omega\ m \qquad \rho_B = (1.2 \pm 0.1) \times 10^{-6}\ \Omega\ m$$

Comparison and discussion

The expected resistivity was (1.00–1.20) × 10^{-6} Ω m. Both parts agreed with this range when uncertainty was included. Part B used the five readings with #30 excluded.

To check whether Parts A and B agreed, we compared the difference between the results with their added uncertainties:

$$|\rho_A - \rho_B| = |1.173764 - 1.206494| \times 10^{-6}\ \Omega\ m = 0.032730 \times 10^{-6}\ \Omega\ m$$

$$\Delta\rho_A + \Delta\rho_B = (0.055487 + 0.110207) \times 10^{-6}\ \Omega\ m = 0.165694 \times 10^{-6}\ \Omega\ m$$

The difference was smaller than the added uncertainties, so Parts A and B agreed within uncertainty.

Part A was almost a straight line. This showed that a longer wire had more resistance, as expected. The line started slightly above zero, which could be due to resistance in the leads or connections.

In Part B, the five included readings followed a nearly straight line, with thinner wires having more resistance. The intercept was close to zero. The #30 reading should be measured again to check whether it was a measurement problem.

Possible sources of error

Poor connections could add extra resistance and make readings too high. If the connections changed between readings, the points could spread out more. We could reduce this by cleaning the contacts and making sure the leads were firmly attached.

The actual wire diameter might be different from the value in the AWG table. This would make the calculated area wrong and affect the resistivity. For example, using too large a diameter in Part A would make the resistivity too high.

Conclusion

Part A showed that longer wire had more resistance and gave ρ = (1.17 ± 0.06) × 10^{-6} Ω m, which agreed with the expected range. Part B gave ρ = (1.2 ± 0.1) × 10^{-6} Ω m with #30 excluded. This agreed with Part A and the expected range within uncertainty.

References

PHYS 210 Lab Manual Complete 2024, Experiment 7, pp. L7-1–L7-4, and Appendix A.

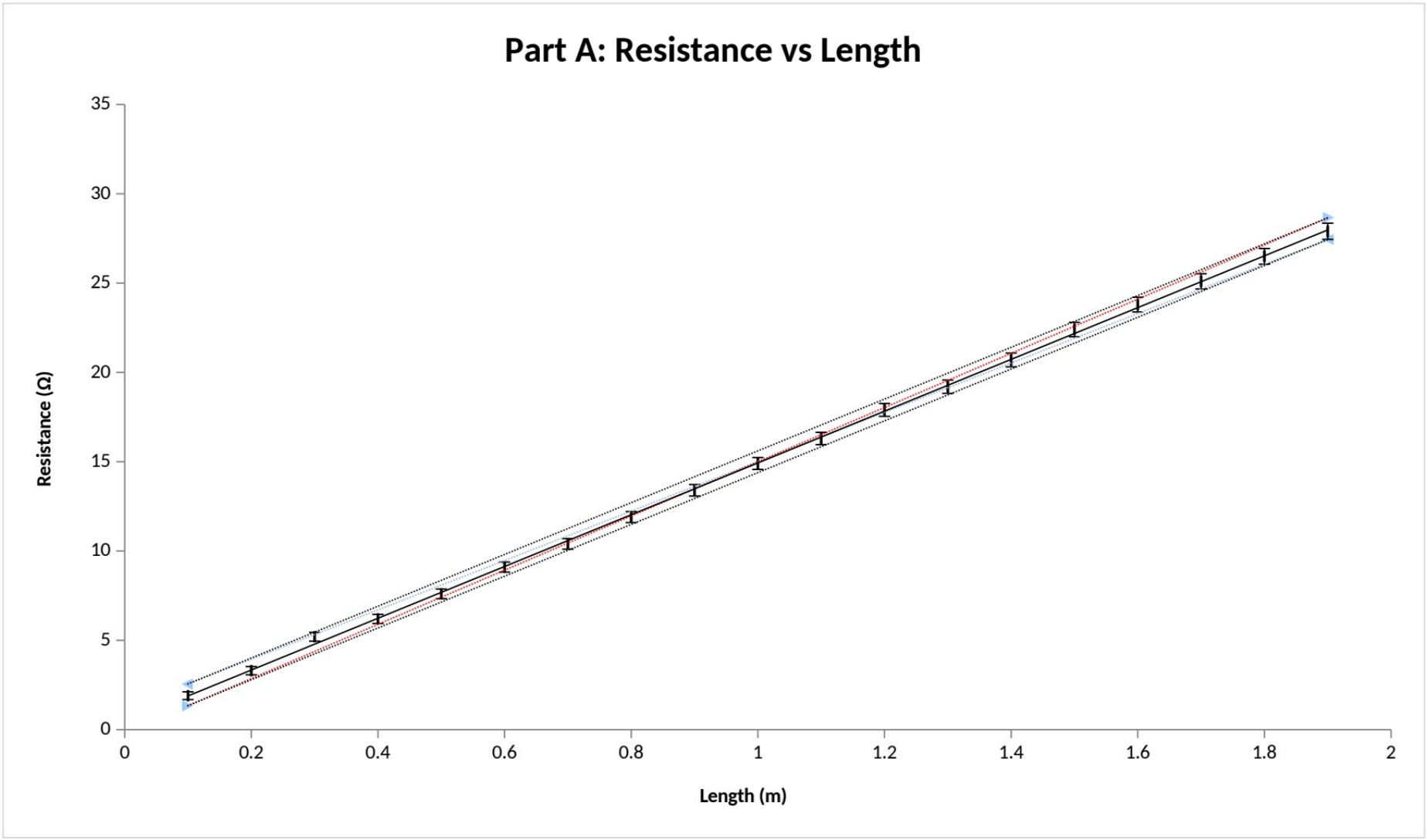


Figure 1. Resistance against length for #28 Nichrome wire.

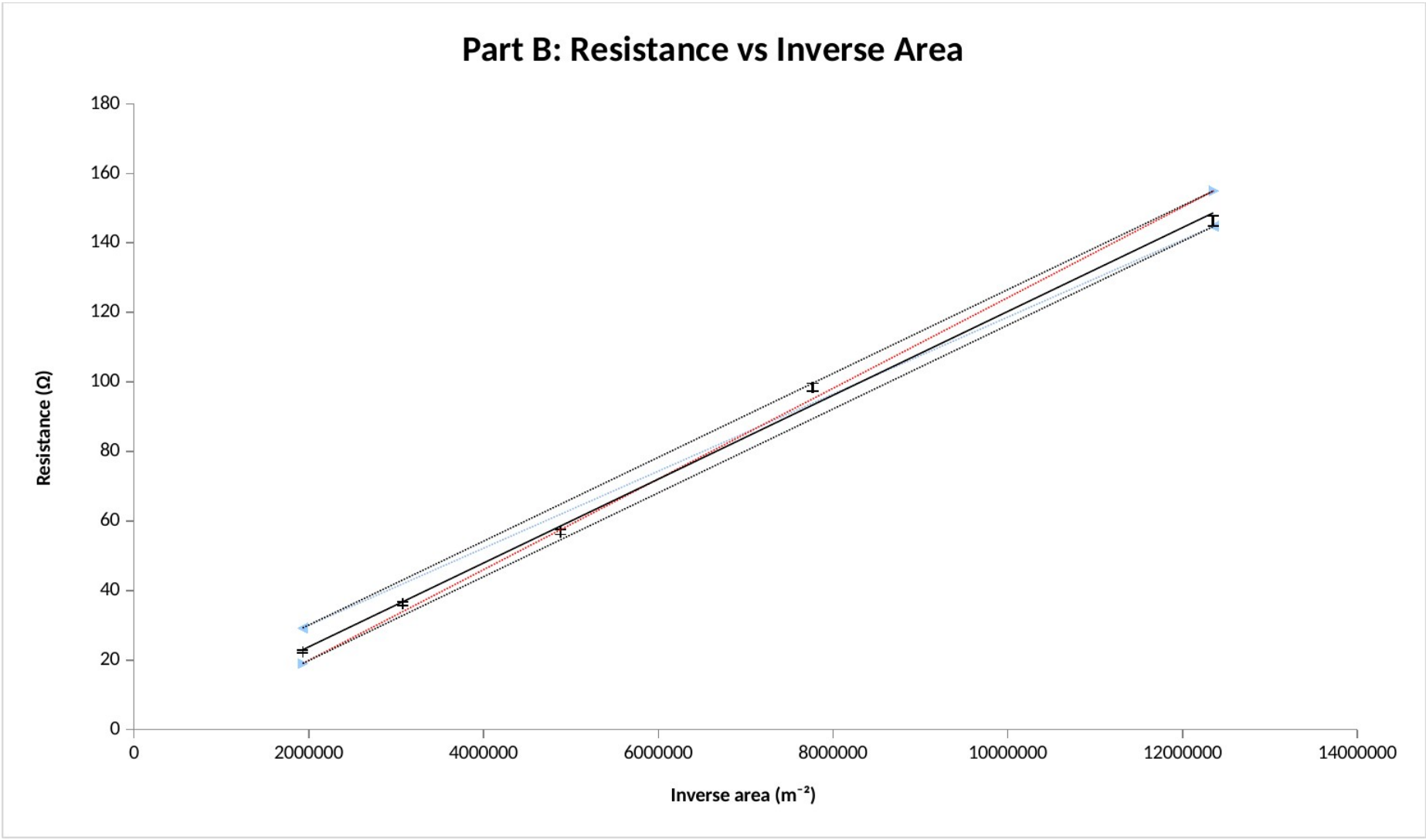


Figure 2. Resistance against inverse cross-sectional area for 10.0 m Nichrome wires (#30 excluded).